

AND THE PLACEBO EFFECT WITHIN THE PREDICTIVE PROCESSING FRAMEWORK

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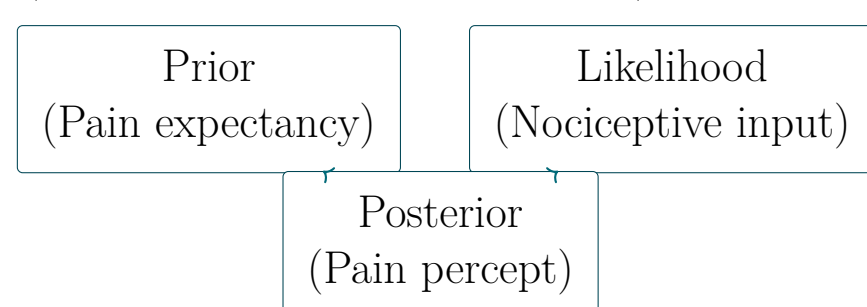
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Background & Motivation

Pain perception is a constructive process in which top-down expectations combine with bottom-up nociceptive input. Within the **predictive-processing** / **Bayesian** framework the final pain percept is the posterior formed by precision-weighted integration of a prior (expected pain) and a likelihood (sensory evidence).



Placebo hypoalgesia is a classic example of expectation-driven pain modulation. Trait **responsiveness to verbal suggestion** (REVS; measured by BSS-C) predicts placebo responding in some paradigms, mediated by expectancy. The latent computational mechanisms that link REVS to placebo hypoalgesia remain unclear.

Research Questions & Hypotheses

Questions

1. Which latent Bayesian parameters best explain individual differences in placebo hypoalgesia?
2. Does the ability to form *precise* pain priors mediate the relationship between REVS and the magnitude of the placebo effect?

Hypotheses

- **H1a** Prior-precision model will be the overall best-fitting model (RFX probability).
- **H1b** A subset of participants will be better explained by the combined prior-shift + prior-precision model.
- **H1c** Average prior precision ($p_{\text{precision}}$) correlates positively with the placebo effect.
- **H2a** REVS (BSS-C) correlates positively with $p_{\text{precision}}$.
- **H2b** $p_{\text{precision}}$ mediates the REVS–placebo association.

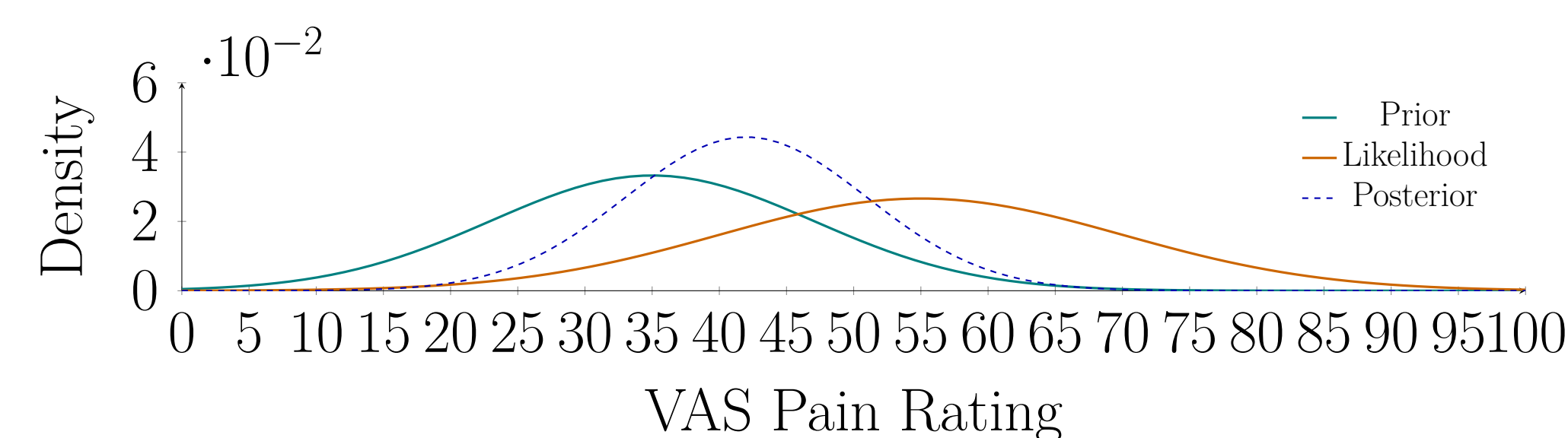
Methods – Experimental Paradigm

Re-analysis of Experiment 2 from Parsons et al. (2021), $N = 78$.

- Double-blind, repeated-measures design.
- Verbal suggestion that a sham cream on one forearm region (Placebo) reduces pain; identical cream on the other region (Control) has no effect.
- 64 trials of calibrated electrical stimulation (identical intensity).
- Trial-by-trial VAS ratings (0–100) of expected pain (prior to stimulus) and experienced pain (after stimulus).
- REVS measured with the Brief Suggestibility Scale (BSS-C); experimenter blind to scores.

Bayesian Computational Models

Pain ratings modelled as Gaussian distributions.
Prior = condition-specific pain-expectancy ratings;
Likelihood = combined pain ratings across conditions;
Posterior = analytic product of the two Gaussians.



Three nested models (VBA toolbox, variational Bayes):

- **Prior-shift model** – free parameter p_{shift} moves the mean of the prior.
- **Prior-precision model** – free parameter $p_{\text{precision}}$ (log change in precision) modulates prior variance.
- **Combined model** – both parameters free.

Model comparison via random-effects (RFX) Bayesian model selection.

Key Results from Exploratory Re-analysis (Experiment 1)

- Prior-precision model best explained **45/57** participants (79%); combined model **12/57** (21%).
- Participants better fit by the combined model showed significantly larger placebo effects.
- Average prior precision correlated positively with placebo magnitude ($r = .46$, $p < .001$).
- REVS did *not* predict placebo responding in Experiment 1 ($r = .11$, ns), consistent with the original report.

These findings motivated the pre-registered re-analysis of the independent Experiment 2 dataset in which REVS *did* predict placebo hypoalgesia.

Planned Analyses (Experiment 2)

- Pearson/Spearman correlations between average $p_{\text{precision}}$, placebo effect (Control – Placebo), and BSS-C.
- Bootstrap comparison of correlation strengths (10 000 resamples).
- Partial correlations controlling for expectancy and other parameters.
- Mediation analysis (SEM + bootstrap) testing whether $p_{\text{precision}}$ mediates the REVS–placebo path.
- Group comparison (Welch t -tests) of placebo magnitude between participants best explained by precision vs. combined models.

Expected Outcomes & Implications

If the prior-precision model again emerges as the dominant explanation and $p_{\text{precision}}$ correlates with both the placebo effect and REVS, the results will indicate that:

- Individual differences in placebo hypoalgesia are largely driven by the *precision* (rather than merely the mean) of pain expectations.
- Trait responsiveness to verbal suggestion may operate, at least in part, by enabling the formation of more precise priors.

This would provide a computationally explicit bridge between a classic individual-difference measure (REVS) and a contemporary predictive-processing account of placebo analgesia.

References & Acknowledgements

Büchel et al. (2014). *Neuron*. Anchisi & Zanon (2015). *PLoS Comput Biol*. Parsons et al. (2021). *Psychosom Med*. Wiedner & Terhune (2019). *Int J Clin Exp Hypn*. Daunizeau et al. (2014). VBA toolbox.

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